



**VAAAL UNIVERSITY
OF TECHNOLOGY**

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Project Report

Faculty	Engineering and Technology
Department	Electrical Engineering
Course	EEEELE2A – Electronics 2
Title	Music Amplifier from Cellphone to Speaker
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Members of this project

The following members contributed to this project:

Student Number	Surname	Initials
240548531	Ngwenya	MB
224367072	Munyai	S

What did each student contribute to this project

Ngwenya MB (240548531) Contribution.

- Audio Input
- Power Control (ON/OFF Switch)
- Input Buffer (Common-Collector Amplifier)

Munyai S (224367072) Contribution

- Audio Amplification
- Volume Control
- Output Stage (Speaker)

Project Report Title

Music Amplifier from Cell Phone to Speaker

Focus of this project

Objective

The objective of this project is to:

- Capture an audio signal from a cell phone.
- Amplify the signal using a transistor amplifier circuit.
- Deliver the amplified signal to a speaker.
- Provide control over the speaker's audio level.

Background

In today's world, amplified audio is essential for activities such as playing music to larger audiences, making announcements, and enhancing communication. However, the output power of a mobile phone or the human voice is limited and insufficient to cover wide areas or drive speakers directly. To overcome this limitation, an amplifier circuit is introduced. An amplifier increases the amplitude of weak audio signals so they can effectively drive a load such as a speaker. The LM386, a low-voltage audio amplifier integrated circuit, is commonly used for such applications because it can deliver adequate gain with minimal external components. Amplification technology plays a vital role in many systems, including radios, public address (PA) systems, walkie-talkies, and home audio equipment.



Figure 1: Transistor Amplifier Application

Components used

The components used in this project include:

- Resistors: 10 k Ω x 2, 330 Ω
- Capacitors: 100 μ F x 2, 1000 μ F x 1, 2.2 μ F
- Power Supply: 9 V Battery
- Output Device: Speaker (0.25 W, 4 Ω)
- Amplification Device: LM386 Audio Amplifier
- Transistor: 2N3904
- Input Interface: 3.5 mm audio jack
- Potentiometer: 5 k Ω , 100 k Ω
- Switch: Single-Pole-Double-Throw (SPDT)
- Board: PCB Board
- Plastic Container
- LED: Green

Software used for circuit drawings and simulation

The design and analysis of the amplifier project were supported by specialized engineering software tools to ensure accuracy, reliability, and efficiency before physical implementation.

- **Multisim:** National Instruments' Multisim was employed for circuit simulation and testing. Its intuitive graphical interface and extensive component library allowed for accurate schematic capture and virtual experimentation with the amplifier. Multisim's real-time simulation capabilities provided valuable insights into the input-output behaviour and overall performance of the circuit, reducing trial-and-error during practical construction.
- **SMath Studio:** This mathematical software was used for modelling and supporting calculations. SMath Studio assisted in deriving component values, performing signal analysis, and verifying theoretical formulas. Its symbolic and numerical computation features ensured that design parameters aligned with the theoretical expectations of the amplifier.

By combining simulation in Multisim with mathematical verification in SMath Studio, the project achieved both practical accuracy and theoretical consistency, leading to a more reliable and optimized amplifier design.

Where will this project affect society in general

Societal Benefits

This music amplifier project from a phone has many societal benefits. The projects benefit the society where a community events, classroom or small gathering takes place and affordable sound amplifier is required to reach ears of each audience. The project's cost a low budget, therefore individual from different background can be able to afford it.

Economic Benefits

The demand for low-cost speakers is high. This project can be scaled to a small business that sells speaker and an affordable price. This business model will not produce high quality low-cost speakers but also give people job opportunity. Individuals from coming from different background could get skills in electronics. All together this project will limit the reliance on expensive commercial audio electronics and reduce electronic waste and energy consumption.

Budget and costing of this project

To construct this project, components were sourced from Swift Electronics, located at: 36 FW Beyers Street, Vanderbijlpark, 1900. Swift Electronics provides affordable electronic components suitable for educational and practical applications. The table below summarizes the components used, their quantities, cost per unit, and total expenditure.

Components	Quantity	Cost Per unit (ZAR)	Total (ZAR)
1 kΩ Resistors	2	Free	0.00
100 μF 25V Capacitor	2	1.00	2.00
1000 μF 25V Capacitor	1	3.50	3.50
2.2 μF 25V Capacitor	1	1.00	1.00
LM386 audio amplifier	1	6.50	6.50
Mini male-to-male 1m aux cable	1	19.50	19.50
9V Battery	2	16.50	33.00
Speaker (8Ω, 0.25W)	1	80.00	80.00
Plastic Container	1	35.00	35.00
PCB Board	1	15.00	15.00
SPDT Switch	1	14.50	14.50
Subtotal			210.00
VAT			2.75
Total			212.75

Table 1: Project Budget and Cost

Circuit diagram or project setup and implementation

Project Setup

The design and construction of the music amplifier were organized into six functional stages, as illustrated in *Figure 2: Block Diagram of the Music Amplifier*. Each stage plays a specific role in processing the audio signal from the mobile phone to the speaker, and they are described as follows:

1. Audio Input Stage

- The audio signal is sourced from a mobile phone through a 3.5 mm auxiliary (AUX) cable connected to the headphone port.
- This weak signal serves as the input to the amplifier system for further processing.

2. Power Control (ON/OFF Switch)

- A simple switch is integrated into the circuit to control power delivery to the amplifier.
- This stage ensures the circuit can be safely powered on or off when required.

3. Input Buffer (Common-Collector Amplifier)

- Configured in a common-collector arrangement, the input buffer provides high input impedance, preventing the mobile phone output from being loaded or attenuated.
- It also delivers a low output impedance, making it suitable to drive the LM386 input stage efficiently.

4. Audio Amplification Stage (LM386 IC)

- The buffered signal is fed into the LM386 audio amplifier integrated circuit.
- At this stage, the weak audio signal is amplified to a sufficient level to drive a loudspeaker without significant distortion.

5. Volume Control Stage

- A potentiometer is connected to the amplifier output as a variable resistor.
- This allows the user to manually adjust the output amplitude, providing control over the loudness of the sound delivered to the speaker.

6. Output Stage (Speaker)

- The amplified signal is delivered to a 4 Ω , 0.25 W speaker.
- In this stage, the electrical signal is converted back into audible sound waves, completing the audio path from the mobile phone to the speaker.

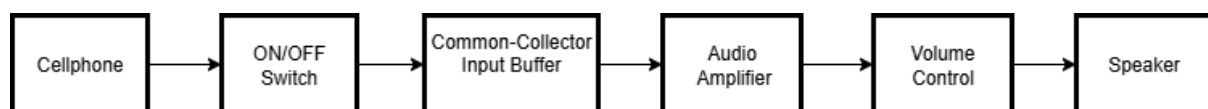


Figure 2: Block Diagram of Music Amplifier

Circuit Diagram

Before physical construction, the circuit was designed and simulated using LTspice software. Simulation was used to verify the performance of the LM386 amplifier, ensure proper biasing, and determine suitable resistor and capacitor values. The simulated schematic is shown in *Figure 3*.

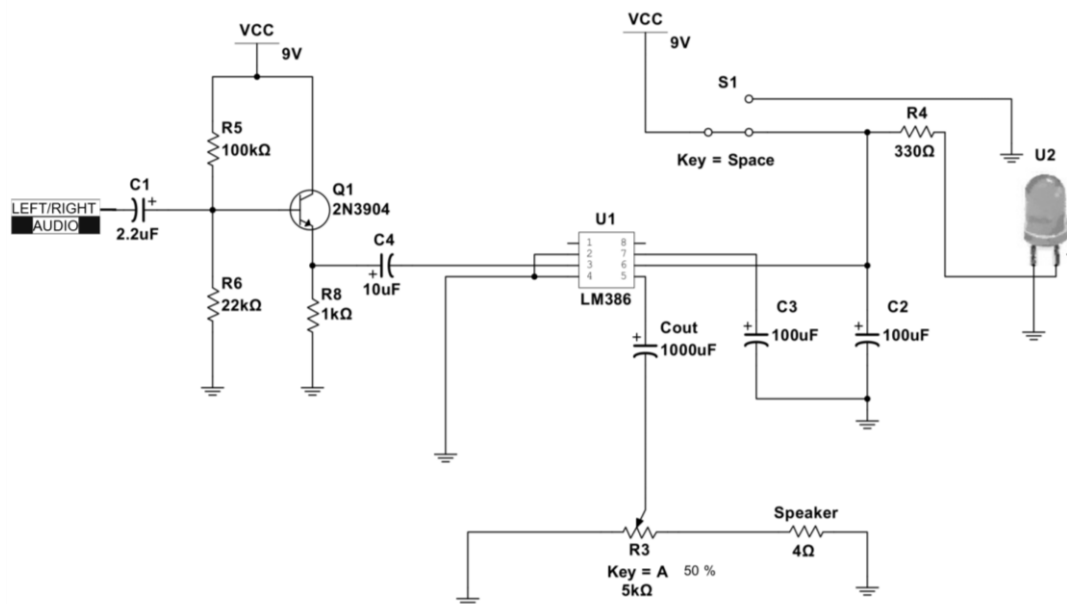


Figure 3: Schematic Diagram of the Circuit

The simulation confirmed that the amplifier could successfully take a small input signal (millivolt range) and increase it to drive the connected speaker.

After successful simulation, the circuit was assembled on a breadboard for initial testing. This stage allowed verification of component connections, observation of signal amplification, and checking for any issues such as overheating or unstable output. The breadboard prototype is shown in Figure 4.

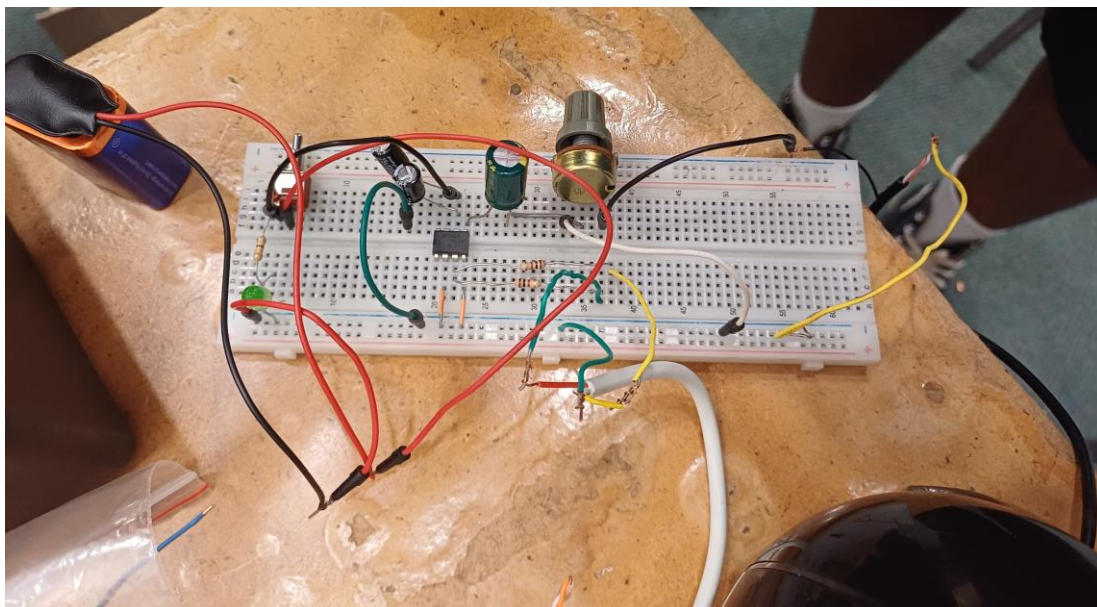


Figure 4: Breadboard Prototype Circuit

Photo of the project

The figures below illustrate the practical construction of the Music Amplifier from Cellphone to Speaker project. These images provide a clear representation of both the internal circuit configuration and the final assembled product. The internal structure of the project is shown in Figure 7.

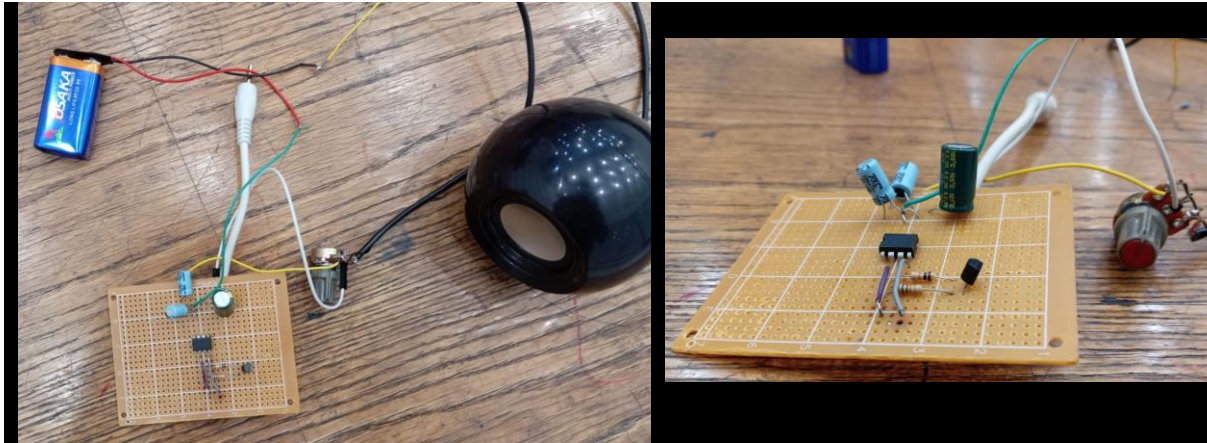


Figure 5: Internal View of the Circuit Mounted on the PCB

The completed project is displayed in Figure 6. In this stage, the PCB and other components were enclosed within a plastic container to provide protection and enhance the project's aesthetic appearance. Only the speaker input cable and volume control knob extend outside the enclosure for easy access. The design ensures safety from accidental electrical contact while maintaining user convenience and portability.



Figure 6: Completed Amplifier Enclosed in Container

Project outcomes

Functional Results

The low signal level produced by music from a cellphone through an AUX cable was insufficient to power a 4 Ω speaker directly. To overcome this, the weak signal was successfully amplified by the common-collector circuit and the audio amplifier, producing sufficient power to drive the speaker. The capacitors used in the circuit effectively reduced high-frequency attenuation and filtered out DC components, resulting in clear and undistorted sound output. The volume control operated smoothly, allowing precise adjustment of the sound level as intended.

Project Objective Fulfilment

The main objective of the project was to capture an audio signal from a cellphone, amplify it using a transistor amplifier circuit, deliver the amplified signal to a speaker, and provide control over the speaker's audio level. All these objectives were successfully achieved. In addition, the project fulfilled academic requirements by demonstrating both theoretical and practical understanding of electronic amplifier design using a transistor and an LM386 audio amplifier.

Calculated values, simulated values and measurements compared

Calculations

The speaker that was used is rated with the power of $P = 3.3\text{ W}$ and a resistance of $R = 4\Omega$. The input voltage of the signal coming from the phone through a cable is determined to have an average voltage of 48.2 mV , and the voltage gain of the LM386 is assumed to be 20. The output values of the speaker if the speaker volume is turned to high the output parameters are determined as follows.

$$V_{in} := 48.2 \text{ mV}$$

$$A_v := 20$$

$$R_{speaker} := 4 \Omega$$

$$V_{out} := A_v \cdot V_{in} = 0.964 \text{ V}$$

$$P_{speaker} := \frac{V_{out}^2}{R_{speaker}} = 0.2323 \text{ W}$$

$$I_{out} := \frac{V_{out}}{R_{speaker}} = 241 \text{ mA}$$

The high-pass cutoff frequency produced in the speaker can be computed as follows

$$R_{speaker} := 4 \Omega$$

$$C_{out} := 1000 \mu\text{F}$$

$$f_c := \frac{1}{2 \cdot \pi \cdot R_{speaker} \cdot C_{out}} = 39.7887 \text{ Hz}$$

Simulated Values

The audio signal coming from the phone constantly change in frequency and the average of the change was determined to be 48.2 mV . The input voltage is then modelled using a signal voltage and the oscilloscope drawing is seen in the below figure with the scale as 50 mV/div .

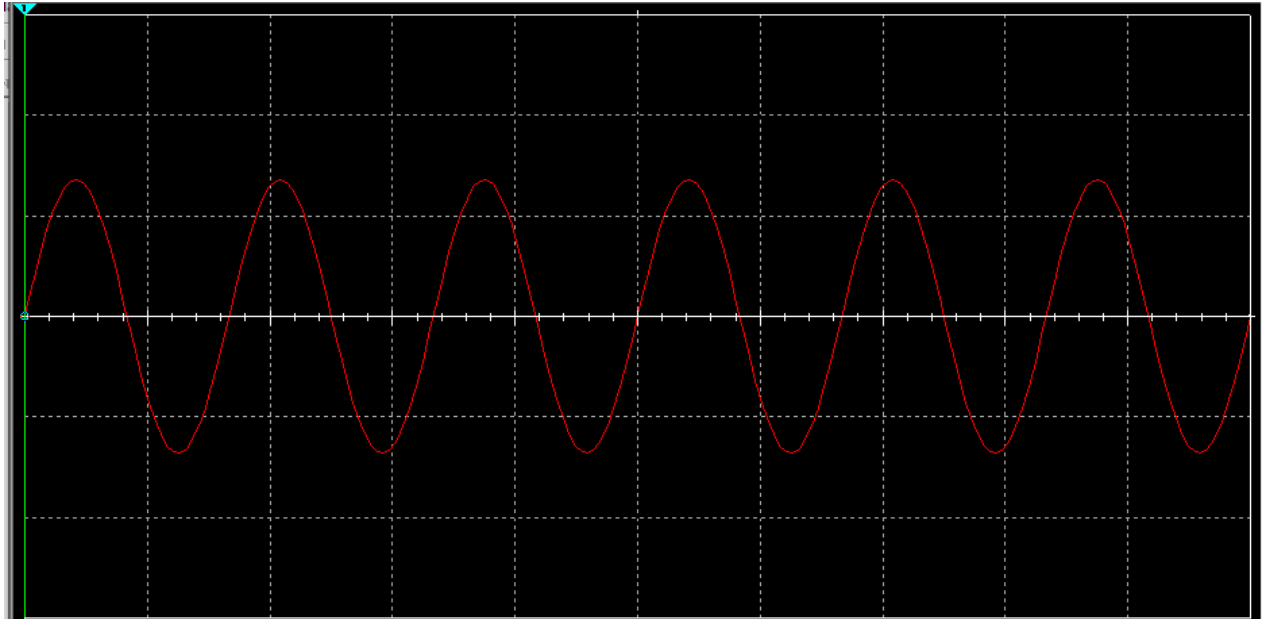


Figure 7: Input Signal Voltage from Phone

The output signal voltage measured across the speaker at full volume is measured with a scale of 1V/div as seen below

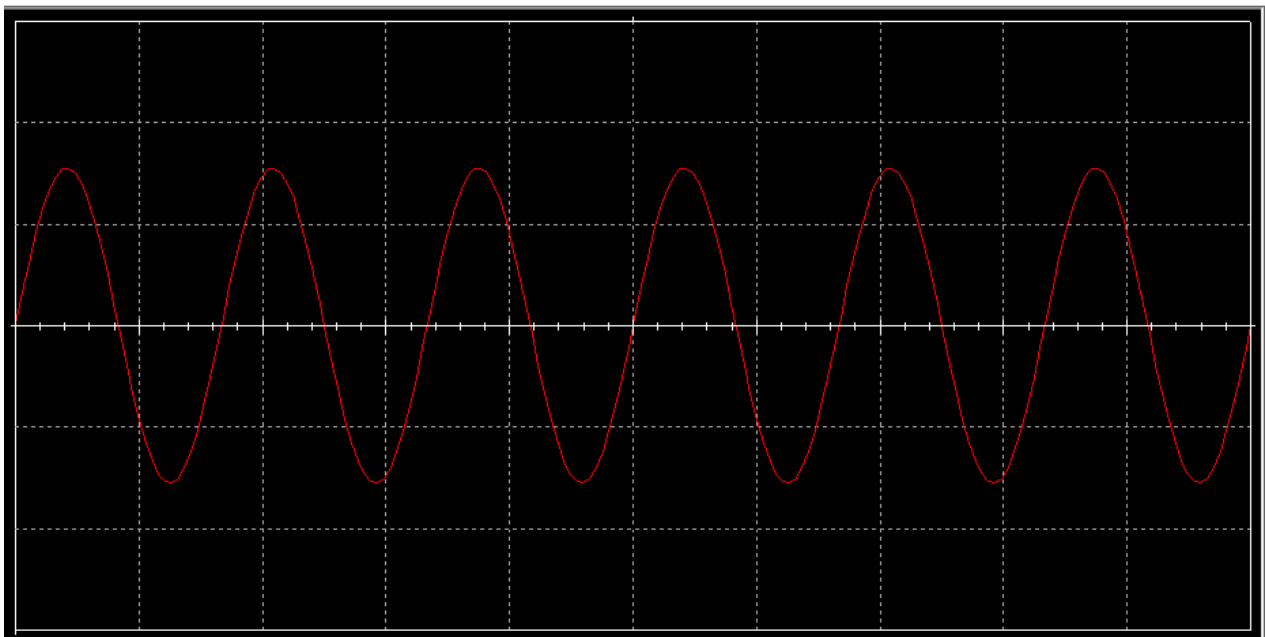


Figure 8: Output Signal Voltage from Speaker

Comparison of Calculated, Simulated and Measured Values

To evaluate the performance and accuracy of the amplifier circuit, calculated, simulated, and measured values were compared. The theoretical (calculated) values were obtained using standard electrical formulas, while simulated results were derived using Multisim. The measured values were recorded from the actual prototype using a multimeter and oscilloscope. This comparison helped verify the reliability of the design, identify minor deviations caused by real-world component tolerances, and confirm that the practical results closely matched the theoretical expectations.

Parameters	Calculated	Simulated	Measured
f_c	19.894 Hz		
X_c	8 Ω		
$V_{in(RMS)}$	48.2 mV	48.2 mV	32.4mV
$V_{speaker(RMS)}$	0.964 V	1.2V	0.76V
$I_{speaker(RMS)}$	241 mA		183.2 mA

Table 2: Comparison of Calculated, Simulated and Measured

What did learn new, What did you not know before completing this project, What challenges did you overcome

What we Learned New

This project provided us with hands-on experience in audio amplifier design using various electronic components. We learned how to use an audio amplifier to process audio signals produced from a cellphone and gained a deeper understanding of how these signals are amplified and delivered to a speaker.

We also developed technical skills during construction, including circuit wiring, soldering, and the proper use of hand tools such as wire cutters and pliers. Additionally, we gained experience in using Multisim for circuit simulation and SMath Studio for performing analytical calculations. Through testing and simulation, we developed a stronger understanding of how signal strength, gain, and distortion affect overall sound quality and amplifier performance.

What We Did Not Know Before Completing this Project

Before building this project, we had limited knowledge of how audio signals differ from standard sine waves. We learned that audio signals constantly vary in frequency and amplitude to produce different sounds, whereas a sine wave has a fixed frequency and amplitude.

We also lacked understanding of how coupling, bypass, and filtering capacitors influence the frequency response of a circuit. During this project, we applied this knowledge — using capacitors to block DC, limit low-frequency gain, and improve amplification at specific frequency ranges. In addition, we had little prior experience working with the LM386 audio amplifier, and this project helped us understand its practical application in amplifying low-level audio signals from a mobile phone.

What Challenges Did We Overcome

When compiling the project, we experienced a couple of problems which include:

- **Component Sourcing:**
Some components were difficult to find locally, requiring us to travel long distances to reach suppliers. In some cases, components did not perform as expected, and we had to return multiple times to exchange or replace them.
- **Circuit Design:**
Our initial design used a multistage push-pull amplifier combining common-emitter and common-collector (emitter-follower) configurations. However, this setup failed to provide

sufficient amplification to power the speaker and resulted in significant power losses due to wiring. Additionally, the circuit layout was too large for practical implementation. To solve this, we redesigned the circuit using a common-collector stage to reduce input impedance, followed by an LM386 audio amplifier to boost the low-level signals from the cellphone effectively.